Solving max.|min. problems

The 2nd Derivative Test (RECALL)

1. Find the critical numbers of $f'(x)$

   (ie: $f'(c) = 0$

2. Substitute the critical numbers into $f''(x)$.

   → If $f''(c) > 0 \Rightarrow$ local minimum at c.

   → If $f''(c) < 0 \Rightarrow$ local maximum at c.

   → If $f''(c) = 0 \Rightarrow$ no conclusion. use chart on $f'(x)$.

Optimization – Numbers

Example 1

Determine two positive numbers with a product of 450 such that the sum of one of the numbers and twice the second number is as small as possible.

$$xy = 450$$
$$y = \frac{450}{x}$$
$$M = x + 2y$$
$$M = x + \frac{900}{x}$$
$$M' = 1 - \frac{900}{x^2}$$
$$1 - \frac{900}{x^2} = 0$$
$$x^2 - 900 = 0$$
$$x = 30$$
$$\Rightarrow y = 15$$
$\therefore$ The 2 #'s are 30 & 15.

$$* \; M'' = \frac{1800}{x^3}$$
$$M''(30) = \frac{1800}{(30)^3} > 0$$
$$\Rightarrow \text{min.}$$

Example 2

The sum of two positive numbers x and y is 150. Determine the two numbers such that the product x^2y is a maximum.

$$x + y = 150$$
$$y = 150 - x$$
$$P = x^2y$$
$$P = x^2(150 - x)$$
$$P = 150x^2 - x^3$$
$$P' = 300x - 3x^2$$
$$-3x(x - 100) = 0$$
$$\Rightarrow x = 0 \;;\; x = 100$$
$x = 0$ inadmissible

$$\therefore y = 50$$
$\therefore$ The 2 #'s are 100 & 50.

$$* \; P'' = 300 - 6x$$
$$P''(100) = -300 < 0$$
$$\Rightarrow \text{max.}$$